

Lesson 4 Teacher's Key

Godot Engine

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

A Hitbox Area2D added to both bullet.tscn and enemy.tscn, with the collision logic living entirely in enemy.tscn's own script:

```
# enemy.tscn's script (attached to the Sprite2D root):
extends Sprite2D

func _ready():
    $Hitbox.area_entered.connect(_on_hitbox_area_entered)

func _on_hitbox_area_entered(area):
    area.get_parent().queue_free()
    queue_free()
```

Level 2 - Complete Worked Solution

queue_free() is safe to call more than once on the same node (a repeat call is simply ignored), so even the rare case of two bullets overlapping the Enemy on the very same frame doesn't cause an error - both bullets are destroyed and the Enemy is destroyed exactly once. No extra code is needed for Level 2.

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on hit, safely

Freeing the Enemy immediately would cut its sound off before it finishes, so hide it and disable its Hitbox right away, play the sound, then await the AudioStreamPlayer2D's finished signal before actually freeing the node:

```
func _on_hitbox_area_entered(area):
    area.get_parent().queue_free()
    hide()
    $Hitbox.set_deferred("monitoring", false)
    $Hit.play()
    await $Hit.finished
    queue_free()
```

Visual: give the enemy its own look

Set modulate in the Enemy's own _ready() so it clearly reads as a threat rather than another shot:

```
func _ready():
    $Hitbox.area_entered.connect(_on_hitbox_area_entered)
    modulate = Color(0.8, 0.1, 0.2)
```

Structure: keep hit-response in enemy.tscn's own script

Everything about how the Enemy responds to being hit - destroying the bullet, hiding itself, playing a sound - lives inside enemy.tscn's own script. bullet.tscn's own script doesn't need to know the Enemy exists at all; it only needed a Hitbox added to it so the Enemy's Area2D could detect it.

Complete Level 3 solution (all 3 goals combined)

Everything above combined, inside enemy.tscn's own script:

```
extends Sprite2D

func _ready():
    $Hitbox.area_entered.connect(_on_hitbox_area_entered)
    modulate = Color(0.8, 0.1, 0.2)

func _on_hitbox_area_entered(area):
    area.get_parent().queue_free()
    hide()
    $Hitbox.set_deferred("monitoring", false)
    $Hit.play()
    await $Hit.finished
    queue_free()
```